TESTING REPORT

Network & Port Scanner

Python Project — Automated Scan Validation

Project	Network & Port Scanner (Python)
Document Type	Testing Report
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Modules Tested	main.py | utils.py | config.py
Test Environment	Own computer (host) → Metasploitable 2 (target VM)

1. Overview

This document describes the test plan, methodology, and results for the Network & Port Scanner Python project. The scanner reads configuration from environment variables, pings each IP in the specified range to confirm it is online, then probes each port in the configured range via TCP connect. Results are written to a CSV file and printed in the console.

The three source files used are:

- config.py — environment loading, validation, and typed config dictionary
- utils.py — host-alive ping detection and individual port scanning
- main.py — orchestration, CSV writing, and console reporting

2. Test Environment

Operating Systems	Own computer (host) → Metasploitable 2 (target VM)
Python Version	3.10
Dependencies	python-dotenv 1.0.1 | Standard library only (socket, subprocess, csv, ipaddress, os, plateform, socket)
Target Hosts	192.168.129.1 – 192.168.129.254 (TARGET_IP_START / TARGET_IP_END) From network 192.168.129.0/24
Port Range Tested	0 – 10000 (START_PORT=0, END_PORT=10000)
Timeout Setting	5 seconds (TIME_OUT_AMOUNT=5)

3. Configuration Tests (config.py)

3.1 Valid Configuration

A .env file was used containing valid IPv4 addresses, integer ports, and a numeric timeout. The load_config() function returned a fully typed dictionary with no errors.

```
START_PORT=20
END_PORT=55
TIME_OUT_AMOUNT=5
TARGET_IP_START=192.168.129.39
TARGET_IP_END=192.168.129.40
```

3.2 Invalid Configuration Scenarios

Scenario	Expected Behaviour	Result
START_PORT > END_PORT	ValueError raised → message printed → exit(1)	**CONFIGURATION ERROR: START_PORT > END_PORT**
Non-integer port value	TypeError caught → exit(1)	**CONFIGURATION ERROR: invalid literal for int() with base 10: 'A'**
Invalid IP address string	AddressValueError caught → exit(1)	**CONFIGURATION ERROR: Octet 390 (> 255) not permitted in '192.168.129.390'** **------** **CONFIGURATION ERROR: Only decimal digits permitted in 'A' in '192.168.129.A'**

4. Host Detection Tests (is_host_alive)

The is_host_alive() function issues a single ping (-c 1 on Linux) with a 500 ms wait. The return code of the ping subprocess determines whether the host is considered ONLINE or OFFLINE.

4.1 Online Hosts

- **192.168.129.39 (Metasploitable 2 VM — ON) — answers to ping : ONLINE, scanning**

4.2 Offline / Unreachable Hosts

- **192.168.129.39 (Metasploitable 2 VM — OFF) — no answers : OFFLINE, not scanning**
- **192.168.129.40 (No host on this address) : OFFLINE, not scanning**

4.3 Cross-Platform Ping Flag

In is_host_alive()

```
command: list[str] = ['ping', param, '1', '-w', '500', ip]
```

Tested from our personal host machine. The -c flag is correctly recognized on Linux. The -w 500 flag may behave differently depending on the distribution (milliseconds or seconds). The behavior observed during testing is correct.

Note: The scan timeout is set to 5 seconds (TIME_OUT_AMOUNT=5). This is a conservative value that correctly catches FILTERED ports via ETIMEDOUT but means each filtered port adds up to 5 seconds to the total scan duration. For 36 ports across 2 hosts in a worst case all-filtered scenario, that is up to 6 minutes. Consider reducing for faster scans on trusted networks.

5. Port Scan Results (scan_port)

The scan_port() function uses a TCP connection (socket.connect_ex) and maps the returned errno code to OPEN, CLOSED, or FILTERED. The table below reproduces the results from the scan_results.csv file obtained during the scan of 192.168.129.39 (Metasploitable 2), ports 20–55, with a 5-second timeout.

5.1 Hôte cible — 192.168.129.39 (Metasploitable 2, ports 20–55)

IP Address	Port	Status	Notes / Service
192.168.129.39	20	CLOSED	FTP
192.168.129.39	21	OPEN	vsftpd 2.3.4 — backdoor connu (CVE-2011-2523)
192.168.129.39	22	OPEN	OpenSSH 4.7p1
192.168.129.39	23	OPEN	Telnet
192.168.129.39	24	CLOSED	Unassigned port
192.168.129.39	25	OPEN	Postfix SMTP — mail service
192.168.129.39	26	CLOSED	Unassigned port
192.168.129.39	27	CLOSED	Unassigned port
192.168.129.39	28	CLOSED	Unassigned port
192.168.129.39	29	CLOSED	Unassigned port
192.168.129.39	30	CLOSED	Unassigned port
192.168.129.39	31	CLOSED	Unassigned port

192.168.129.39	32	**CLOSED**	Unassigned port
192.168.129.39	33	**CLOSED**	Unassigned port
192.168.129.39	34	**CLOSED**	Unassigned port
192.168.129.39	35	**CLOSED**	Unassigned port
192.168.129.39	36	**CLOSED**	Unassigned port
192.168.129.39	37	**CLOSED**	Unassigned port
192.168.129.39	38	**CLOSED**	Unassigned port
192.168.129.39	39	**CLOSED**	Unassigned port
192.168.129.39	40	**CLOSED**	Unassigned port
192.168.129.39	41	**CLOSED**	Unassigned port
192.168.129.39	42	**CLOSED**	Unassigned port
192.168.129.39	43	**CLOSED**	Unassigned port
192.168.129.39	44	**CLOSED**	Unassigned port
192.168.129.39	45	**CLOSED**	Unassigned port
192.168.129.39	46	**CLOSED**	Unassigned port
192.168.129.39	47	**CLOSED**	Unassigned port
192.168.129.39	48	**CLOSED**	Unassigned port
192.168.129.39	49	**CLOSED**	Unassigned port
192.168.129.39	50	**CLOSED**	Unassigned port
192.168.129.39	51	**CLOSED**	Unassigned port
192.168.129.39	52	**CLOSED**	Unassigned port
192.168.129.39	53	**OPEN**	BIND DNS
192.168.129.39	54	**CLOSED**	Unassigned port
192.168.129.39	55	**CLOSED**	Unassigned port

Summary: 5 OPEN ports (21, 22, 23, 25, 53), 31 CLOSED ports, 0 FILTERED ports in this range. The absence of FILTERED ports indicates that Metasploitable 2 actively responds (RST) rather than silently ignoring connections.

5.2 Offline Host 192.168.129.40 — no result

The host 192.168.129.40 is included in the configured range (TARGET_IP_END=192.168.129.40) but did not return any entry in scan_results.csv. This means that is_host_alive() returned False for this IP — no VM is assigned to this address in the virtual network. The scan was correctly skipped.

```
[→] 192.168.129.40 est OFFLINE
```

6. CSV Output Validation

The scan_results.csv file was inspected after a full scan run (192.168.129.39, ports 1–1024).

6.1 File Structure

- Header row present: IP Address, Port, Status
- One data row per port per online host
- No rows written for offline hosts
- Encoding: UTF-8 with no BOM (default csv module)
- Encodage : UTF-8, fins de ligne Linux (\n)

6.2 Sample CSV Output

```
IP Address,Port,Status
192.168.129.39,20,CLOSED
192.168.129.39,21,OPEN
192.168.129.39,22,OPEN
192.168.129.39,23,OPEN
192.168.129.39,24,CLOSED
192.168.129.39,25,OPEN
```

7. Edge Cases

Scenario	Observed Behaviour	Verdict
Single IP range (start == end)	Exactly one host scanned — only one network block iterated	✅ **PASS**
Single port (start == end)	One port evaluated per host — CSV has one row per online host	✅ **PASS**

Large range (0.0.0.0–255.255.255.255)	Valid but impractical — would run for a long time; no guard in config	⚠ WARN

8. Findings & Recommendations

8.1 Issues Found

1. **Ping Timeout Flag (-w):** On Linux, -w is in seconds on some distributions and milliseconds on others. Replace -w 500 with -W 1 for consistent 1-second timeout across all POSIX systems.
2. **No IP Range Guard:** There is no maximum range limit. A misconfigured .env could trigger a scan of millions of IPs.
3. **Host Discovery Latency:** On low-performance hardware, a short timeout can be unefficient. This can cause the tool to misidentify active hosts as "OFFLINE," resulting in empty scan reports. Increasing the `TIME_OUT_AMOUNT` to 5 seconds was necessary to allow the system enough time to process network packets and confirm host availability.

8.2 What Works Well

- Clean separation of concerns across three modules
- Typed config dictionary provides early failure on bad input
- FILTERED vs CLOSED distinction using errno codes
- CSV output is clean
- Offline hosts are skipped efficiently
- Console output provides real-time progress visibility